PROJECT CHARTER

PA2601 Reachability Simulation of Robotic Arm

CLIENT – DR Rhys Tague

Date: 27/03/2026

COMP 3018 Professional Experience

Team:	
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Content

Executive Summary

Throughout the 20th and the 21st century, the development, understanding and evolution of robotics has undergone a massive change. From massive, automated factories employing multi-ton arms capable of lifting a thousand kilograms, to helpful household robots such as Roombas, to the eventual attempted creation of mechanical humanoids, robotics has spread its cogs to nearly every industry available on this planet, including medical, health and patient care through the use of assistive robotics. And throughout the history of developing robotics, researching, measuring, testing and prototyping have always been side-by-side while implementing these new technologies. Especially so in said healthcare fields as the involvement of human life becomes a major factor when developing and utilising these automated androids as the goal of patient-care robots is to help those in need, aiding their lives, guiding them towards a better tomorrow. On the other hand, due to the nature of robotic testing and prototyping, the financial costs and time spent quickly escalate as there are many different scenarios that researchers have to account for within patient care scenarios. Installation of robots, implementation of both software and hardware and repeated data collections all slowly add up to the rising costs associated with such research.

As such, project PA2601 aims to cure this ailment via accurate simulation of real-world spaces replicated within a 3D environment with the use of robotic arms and physics-based simulators. The aim is to collect data over a vast array of repeated simulations over varying differing scenarios aiming to maximise researcher, operator and patient safety while minimising real world costs in testing, hazards and risks.

As mentioned above, given the multiple scenarios, most of our major milestones will be laid out in sequence from most important to least important following the client's needs and wants with each milestone ending with some form of data collected from each of the simulated scenarios. Some milestones that will be added are Bed Ends Reach, Unreachable Goal Handling and Robot Base Placement.

As project PA2601 is set to last 14 weeks, the total net benefit of running this project will be around $25,000 with a cost-benefit ratio of 2.27 or a net return of $1.27 per dollar. Furthermore, as the PA2601 team is still in its onboarding phase, the actual cost of this project is expected to be $0, and as such a net benefit or return should be around $44,600. This is further shown as the resources used during this project term will all be open sourced, meaning there are no costs in terms of software development or licensing fees to be involved.

Project Purpose & Scope

Client Details:

Our client, Dr Rhys Tague (R.Tague@westernsydney.edu.au) is an associate lecturer of computing in Western Sydney University and a Doctor in Philosophy, whose current project goals is the viability of deploying robotic arms in patient care scenarios with the initial task of creating a prototype using a physics-based simulation where data of scenarios and measurements of the robotic arm can be repeatedly gathered through various different user scenarios.

Problem Statement:

Though the creation and development of robotics has come a long way since its dawn, the collection, testing and analysing of robotics in industries, especially in industries involving human life, are still very time consuming and costly, reaching from tens of thousands to the heights of millions depending on use and application. Robotic testing and prototyping scenarios for researchers testing and gathering data and measurements will quickly become expensive, reaching tens of thousands to even hundreds of thousands of dollars, time consuming as each test must be repeated multiple times to confirm performance metrics and correct data, and health risks presented towards robotic-arm operators, test/real patients and researchers present.

And while robotic arm simulators solve many of the financial burdens and personnel's health risks, finding suitable and reliable simulators with specific scenarios demanded from real world environments is like trying to find a needle within a haystack. As such, one may consider the development of a specific program or software to meet their demands, however, it will quickly become an ouroboric problem as the development of such technologies in time, become costly and time consuming. The hiring of a professional development team or individual/s will undoubtedly raise the budget as well as the time needed to complete the development of such software.

Objectives:

The main objective of our project group, as defined by our client, is to create a simulation of various scenarios where the focus of each simulation is to record, collect and measure the data from each of the scenarios. This simulation will act as a prototype and a springboard for the next major project that we will be handing off.

There are 8 given scenarios, divided into 3 levels of difficulties. The Core scenarios, the Sub-Core scenarios and the Side Scenarios.

- The Core Scenarios are the highest priority tasks/milestone that are expected to be completed by the end of the projects life and consists of; Patient Engagement Envelope, Bed Ends Reach, Unreachable Goal Handling, Robot Base Placement Comparison and Time Pressure vs Safety Trade-Off
- The Sub-Core Scenarios are the next highest priority or medium priority where the project should aim to complete these scenarios by the end of the time period and include; Dynamic Goal update and Arm-Region without Collision.
- Side Scenarios are the lowest priority goals, where achieving this goal may not be possible within the given time period, due to reasons such as team skills, time constraints, etc, and includes the task Reinforcement Learning.

Through these 6-8 scenarios, the main focus will be collecting data through these simulations, and measurements taken via things like, mapping (heatmaps or vertical reach maps), speed-limit studies, and more.

Roles, Responsibilities & Team Capability

Team Roles

The PA2601 project team consists of four members, each contributing to different aspects of the simulation-based evaluation system. Roles are assigned based on individual strengths while maintaining strong collaboration across all components.

1. Seaniya Rajalingam – Project Management & Planning

- Responsible for overall project coordination and planning.
- Develops and manages the project timeline, milestones, and deliverables.
- Ensures that all tasks are completed within deadlines and aligned with project objectives.
- Identifies and manages risks and constraints that may impact progress.
- Coordinates communication between team members, supervisors, and clients.
- Monitors team performance and ensure smooth workflow across all modules.
- Supports documentation, report structure, and presentation preparation.

2. Jonathan Ng – System Architecture & Reporting Lead

- Designs the overall evaluation of harness architecture.
- Implements the logging, metrics, and reporting pipeline.
- Generates outputs such as heatmaps, CSV/JSON logs, and performance reports.
- Ensure system scalability, consistency, and data accuracy.

3. Ahnaf Abid Zarif – Risk & Communication Management

- Handles risk assessment and mitigation strategies.
- Develops and maintains the communication plan.
- Coordinates meetings and ensures stakeholder engagement.
- Ensures proper documentation of project risks and updates.

4. Shaurya Garg – Control Systems & Safety Logic

- Develops robot control mechanisms using Inverse Kinematics (IK).

- Implements safety features such as collision detection and safe distance constraints.
- Works on scenarios like unreachable goal handling (S4).
- Ensures robot behaviour complies with safety requirements.

Team Responsibilities

Core Responsibilities

- Develop a simulation environment representing patient-care settings.
- Implement key evaluation scenarios (S1, S2, S4, S7, S8).
- Build a unified evaluation harness with scenario runner, logging, metrics pipeline, and reporting tools.
- Ensure safety-first design (collisions treated as failures, minimum distance enforced).
- Maintain repeatability using deterministic simulation setups.
- Collaborative Responsibilities
- Integration of different modules into a single system.
- Testing and debugging simulation scenarios.
- Documentation and report writing.
- Preparing presentations and demonstrations

Team Capabilities

Technical Capabilities

- Programming: Python for simulation and system development.
- Simulation Tools: Experience with PyBullet.
- Robotics Knowledge: Inverse Kinematics, motion planning, safety constraints.
- Data Analysis & Visualization: Heatmaps, performance metrics, Matplotlib.
- System Design: Modular architecture and scalable evaluation pipeline.

Problem-Solving Capabilities

- Translating real-world patient-care challenges into simulation scenarios.
- Designing safe and efficient robotic behaviors.
- Handling edge cases such as unreachable targets and safety violations.

Teamwork & Communication

- Strong collaboration across roles.
- Effective communication with supervisor and client.

- Agile and iterative development approach.

Project Strengths

- Simulation-based evaluation reduces cost and risk.
- Ability to generate data-driven insights for robotic safety and performance.
- Alignment with real-world healthcare applications.

Summary

The PA2601 team combines expertise in project management, system design, robotics, and simulation. With clearly defined roles and shared responsibilities, the team is well-equipped to deliver a reliable and scalable assistive robotics evaluation system that prioritizes safety, efficiency, and real-world applicability.

Solution Overview

A comprehensive PyBullet-based assessment harness for assistive robots in aged-care settings is provided by the project. Before any real-world deployment, it allows robot researchers and caregivers to safely evaluate robot performance and safety in a completely virtual patient-care environment.

To ensure that the same scenario logic operates consistently, the harness is constructed around a small, simulator adapting layer (with common methods for reset, step, state retrieval, and event application). The main backend is PyBullet, which offers quick, deterministic physics simulation of a basic patient proxy, named room zones, set robot base locations, and precise target poses.

Every test follows strict, repeatable rules: success is measured by end-effector distance and dwell time; safety violations are hard failures; and deterministic seeds guarantee identical results across batch runs.

The harness executes the full recommended core scope:

- S1 (vertical reach envelope / heatmap)
- S2 (head vs foot zone comparison)
- S4 (unreachable goal handling)
- S7 (robot base placement trade-offs)
- S8 (speed vs safety trade-off)

Furthermore, the shared runner, logging, metrics pipeline, and reporting tools. One medium extension (either S3 dynamic goal update or S5 arm-region reach) will also be implemented. The extension can be decided during the demonstration of prototypes or during the iterations.

Robot researchers may utilize the outputs—standardized tables, heatmaps, and summary statistics—to provide evidence-based suggestions, while advocates of elderly care receive precise instructions on safe bedside robot settings that enhance patient movement without sacrificing safety.

Cost Benefit Analysis

COST

Category	Description	Cost (AUD)
Development effort	4 junior developers × approx 140 hours (14 weeks * 10 hrs per week) each valued at $35/hr	$19,600
Infrastructure	PyBullet (open-source) running locally on team laptops	$0
Training/Onboarding	Team learning PyBullet physics, IK, and scenario scripting (using open sources to learn and scenario provided by sponsor)	$0
Reporting tools	Matplotlib/Seaborn for heatmaps and trade-off plots and CSV packages for artifacts	$0
Total Costs		**$19,600**

BENEFIT

Category	Description	Benefit (AUD)
Hardware cost avoidance	Avoids purchase of physical robot arm and bed setup for testing. A full size bed installation and gettting a designated room for it is hard in a full working environment	$10,000
Researcher time savings	Researchers save approx 10 hrs/week on manual testing & analysis. Dont have spend resources in testing with robots	$7,000
Safety & risk reduction	Early identification of unsafe placements and speeds prevents real-world incidents such as patient hazards or physical damage. All	$3,000
Research & deployment value	Publishable metrics and placement recommendations for aged-care facilities	$5,000
Offset (Part of University Course)	This project is part of the University Course. Hence, the students don't get paid for their working hours in the form of money but get course credits on completion of Project	$19,600
Total Benefits		**$44,600**

Net Benefit = $44,600 – $19,600 = $25,000

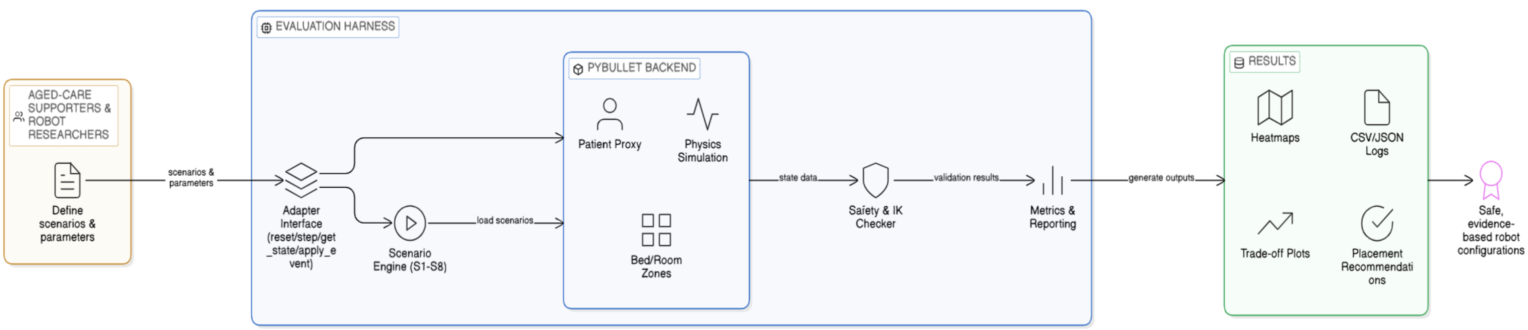
Cost-Benefit Ratio = 2.27 (every $1 invested returns $2.27 in value. A net benefit of $1.27).

Rationale for Chosen Solution

Assistive robot testing in close proximity to actual patients is costly, time-consuming, and poses serious safety hazards. The team gets flawless repeatability, minimal hardware risk, and the capacity to execute hundreds of batch tests overnight by utilizing PyBullet simulation with the precise standardized setup and scenario catalogs. Rather of using one-time demonstrations, the scenario-driven technique generates directly comparable, publishable measurements for reachability, safety margins, base location, and speed limitations. This immediately aids robot researchers in producing suggestions based on evidence and enables supporters of aged care to confidently adopt robots that improve patient mobility while safeguarding both patients and caregivers. Additionally, the harness is future-proofed for any upcoming simulator improvements thanks to the modular adapter design.

High Level Architecture

Workflow Diagram



Requirements

High Level Business Function

ID	High-Level Business Function	Ranking
BF1	Quantify safe robot interaction workspace for aged-care patients	Essential
BF2	Compare robot reachability and performance at head vs foot zones	Essential
BF3	Detect and safely reject unreachable or unsafe goals	Essential
BF4	Evaluate robot base placement trade-offs around the patient bed	Essential
BF5	Measure time-pressure vs safety-margin trade-offs	Essential
BF6	Handle dynamic patient or target shifts without collision	Desirable

Epics and Feature

ID	Type	Name	Linked Business Function	Description
E1	Epic	Reachability Analysis	BF1, BF2	Understand where the robot can safely and effectively reach around the patient
F001	Feature	Patient Engagement Envelope	BF1	Show which areas above the patient the robot can safely reach
F002	Feature	Bed-Zone Reachability Comparison	BF2	Compare how well the robot performs near the head vs foot of the bed

ID	Type	Name	Linked Business Function	Description
E2	Epic	Safety & Failure Handling	BF3, BF6	Ensure the robot avoids unsafe actions and handles problems safely
F003	Feature	Unreachable Goal Handling	BF3	Detect when a task is unsafe or impossible and

				stop early with a clear message
F008	Feature	Arm-Region Reach without Collision (extension)	BF6	Allow safe robot movement near patient arms without collisions

ID	Type	Name	Linked Business Function	Description
E3	Epic	Robot Performance & Optimization	BF4, BF5	Evaluate different robot setups to improve performance and safety
F004	Feature	Robot Base Placement Trade-off Study	BF4	Test different robot positions around the bed and recommend the best one
F005	Feature	Speed vs Safety Trade-off Study	BF5	Compare different robot speeds to balance efficiency and safety

ID	Type	Name	Linked Business Function	Description
E4	Epic	Evaluation System & Reporting	BF1–BF5	Provide a system to run tests and present results clearly
F006	Feature	Evaluation Harness Core	BF1–BF5	Run tests, store results, and generate simple reports for users

ID	Type	Name	Linked Business Function	Description
E5	Epic	Adaptive Robot Behaviour	BF6	Enable the robot to adjust to changes in the environment
F007	Feature	Dynamic Goal Update (extension)	BF6	Allow the robot to adjust safely when the patient or target moves

Prioritized User List

ID: F001	Feature: Patient Engagement Envelope	Priority: Required
BF1: Quantify safe robot interaction		Estimate: 8 hours
Story: As a caretaker, I want to see which areas above the patient the robot can safely reach, so that I can avoid unsafe interactions.		

ID: F001	Feature: Patient Engagement Envelope	Priority: Required
BF1: Quantify safe robot interaction		Estimate: 9 hours
Story: As a nurse, I want a simple visual showing safe and unsafe zones, so that I can quickly understand where the robot can operate.		

ID: F001	Feature: Patient Engagement Envelope	Priority: Required
BF1: Quantify safe robot interaction		Estimate: 8 hours
Story: As a researcher, I want to avoid placing objects in unreachable areas, so that tasks are completed efficiently.		

ID: F002	Feature: Bed-Zone Reachability Comparison	Priority: Required
BF2: Compare robot reachability and performance at head vs foot zones		Estimate: 7 hours
Story: As a caretaker, I want to know whether the robot works better at the head or foot of the bed, so that I can position it correctly.		

ID: F002	Feature: Bed-Zone Reachability Comparison	Priority: Required
BF2: Compare robot reachability and performance at head vs foot zones		Estimate: 6 hours
Story: As a nurse, I want to compare task completion times in different zones, so that I can choose the most efficient setup.		

ID: F003	Feature: Unreachable Goal Handling	Priority: Required
BF3: Detect and safely reject unreachable or unsafe goals		Estimate: 6 hours
Story: As a caretaker, I want the robot to tell me when it cannot reach something, so that I can adjust the task.		

ID: F003	Feature: Unreachable Goal Handling	Priority: Required
BF3: Detect and safely reject unreachable or unsafe goals		Estimate: 6 hours
Story: As a researcher, I want a simple explanation when a task fails, so that I understand what went wrong.		

ID: F004	Feature: Robot Base Placement Trade-off Study	Priority: Required
BF4: Evaluate robot base placement trade-offs around the patient bed		Estimate: 8 hours
Story: As a caretaker, I want to try different robot positions around the bed, so that I can find the safest setup.		

ID: F004	Feature: Robot Base Placement Trade-off Study	Priority: Required
BF4: Evaluate robot base placement trade-offs around the patient bed		Estimate: 5 hours
Story: As a researcher, I want a recommendation for robot placement, so I don't need to test it myself.		

ID: F005	Feature: Speed vs Safety Trade-off Study	Priority: Required
BF5: Measure time-pressure vs safety-margin trade-offs		Estimate: 7 hours
Story: As a caretaker, I want the robot to move at a safe speed, so that it does not harm the patient.		

ID: F005	Feature: Speed vs Safety Trade-off Study	Priority: Required
BF5: Measure time-pressure vs safety-margin trade-offs		Estimate: 8 hours
Story: As a nurse, I want to compare slow and fast movements, so that I can balance speed and safety.		

ID: F006	Feature: Evaluation Harness Core	Priority: Required
BF1 – BF5: Multiple Business Functions		Estimate: 10 hours
Story: As a researcher, I want simple reports after each test, so that I can understand robot performance.		

ID: F007	Feature: Dynamic Goal Update	Priority: Desirable
BF6: Handle dynamic patient or target shifts without collision		Estimate: 5 hours
Story: As a caretaker, I want the robot to adjust when the patient moves slightly, so that the task can continue safely.		

ID: F008	Feature: Arm-Region Reach without Collision	Priority: Desirable
BF6: Handle dynamic patient or target shifts without collision		Estimate: 8 hours
Story: As a caretaker, I want the robot to work safely near the patient’s arms, so that I can use it for close tasks.		

MILESTONE & TIMELINE

Milestone

Week	Milestone	Description
Week 1–2	Project Initiation	Team formation, environment setup, confirm tools (PyBullet/Assistive Gym), define patient proxy, bed layout, robot base pose, safety rules.
Week 3	Evaluation Harness v1	Implement adapter interface (reset, step, get_state, apply_event). Begin logging + metrics pipeline.
Week 4	Project Charter Due (10%)	Submit Project Charter + Sponsor Confirmation by Friday 11:59 pm.
Week 4–5	Scenario S1 Complete	Vertical reach map + heatmap for ≥200 points.
Week 5	Scenario S2 Complete	Head vs foot reach comparison + reporting pipeline.
Week 6 (Mon/Tue)	Prototype Workshop (10%)	Mandatory 1hour workshop with sponsor & supervisor. Demonstrate S1 + S2 + harness.
Week 6–7	Scenario S4 Complete	Unreachable goal handling (early reject vs late failure).
Week 8	Scenario S7 Complete	Base placement comparison (≥30 target points per placement).
Week 9	Iteration One Demo (10%)	Show S1, S2, S4, S7 progress + feedback by Friday 11:59 pm.
Week 9–10	Scenario S8 Complete	Time pressure vs safety trade-off (speed-limit study).
Week 10–11	Extension Scenario	Choose S3 (dynamic goal update) or S5 (arm-region reach).
Week 12	Iteration Two Demo (10%)	Show extension scenario + improved harness by Friday 11:59 pm.
Week 12–13	Final Integration	Combine all scenarios into unified reporting system.
Week 14	Final System Due (20%)	Submit final system + sponsor confirmation by Friday 11:59 pm.
Week 14	Abstract + Video Due (5%)	Submit by Friday 11:59 pm.
Week 14	Handover & Completion Report (5%)	Submit by Friday 11:59 pm.
Week 15	Final Presentation (10%)	Present to coordinator, supervisor, and possibly research team.

Release Scheule Date

ID	Feature / Scenario	Responsible	Iteration	Start Week	Finish Week	Status
F006	Evaluation Harness Core		1	1	3	Planned
F001	Patient Engagement Envelope (S1)	Shaurya Garg	1	4	5	Planned
F002	Bed-Zone Reachability Comparison (S2)	Ahnaf Zarif	1	5	5	Planned
F003	Unreachable Goal Handling (S4)	Seaniya Rajalingam	2	6	7	Planned
F004	Robot Base Placement Trade-off (S7)	Jonathon Ng	2	8	8	Planned
F005	Speed vs Safety Trade-off (S8)	Shaurya Garg and Jonathon Ng	2	9	10	Planned
F007	Dynamic Goal Update (S3) / Extension	All Team Members	3	10	11	Planned
F008	Arm-Region Reach Without Collision (S5)	All Team Members	3	10	11	Planned
F006	Final Integration / Reporting System	All Team Members	3	12	13	Planned
-	Project Charter Submission	All Team Members	1	4	4	Planned
-	Prototype Workshop	All Team Members	1	6	6	Planned
-	Iteration One Demo	All Team Members	2	9	9	Planned
-	Iteration Two Demo	All Team Members	3	12	12	Planned

-	Final System Due	All Team Members	3	14	14	Planned
-	Handover & Completion Report	All Team Members	3	14	14	Planned
-	Final Presentation	All Team Members	3	15	15	Planned

RISK & CONSTRAINTS

RISKS

Risk	Description	Mitigation
Client unavailable in Week 6	Client is on sick leave during Week 6 except Mon/Tue.	Schedule Prototype Workshop Early Week 6.
Complexity of simulation	IK, collision checking, safety thresholds may cause failures.	Build minimal environment first; test each scenario independently.
Batch runner instability	Crashes during long runs may corrupt metrics.	Implement robust logging, checkpointing, deterministic seeds.
Scenario workload too large	Core + extension scenarios may exceed time.	Follow recommended scope: S1, S2, S4, S7, S8 + one extension.
Team skill differences	Varying experience with PyBullet/Assistive Gym.	Pair programming, task rotation, supervisor guidance.
Late integration issues	Scenarios may work individually but fail together.	Integrate early (Week 8–9), test continuously.
Prototype Workshop failure	Poor demo may affect feedback loop.	Ensure S1 + S2 + harness are stable by Week 6.

Overall, these risks highlight the technical, organizational, and scheduling challenges inherent in developing a simulation-based evaluation harness within a fixed semester timeline. By identifying these issues early and applying structured mitigation strategies, the team can maintain steady progress, reduce uncertainty, and ensure that core deliverables remain achievable. Proactive communication with the supervisor and client will be essential to managing these risks effectively throughout the project.

CONSTRAINTS

Constraint	Source	Impact
Standardised simulation setup required	Scenario Catalogue	Limits modelling complexity; ensures comparability.
Simulator agnostic design	Scenario Catalogue	Must implement adapter interface (reset, step, get_state, apply_event).
Collisions = hard failures	Scenario Catalogue	Safety logic must be robust and early.
Fixed assessment deadlines	Assessment Table	Timeline must align with Weeks 4, 6, 9, 12, 14, 15.
No real robot access	Project Brief	All testing must be virtual.
15week semester	Course structure	Requires strict milestone planning.

These constraints define the boundaries within which the project must operate, ensuring consistency, safety, and alignment with academic requirements. While they limit certain design freedoms, they also provide a stable framework that supports comparability across scenarios and adherence to assessment deadlines. Understanding and respecting these constraints will help the team deliver a robust, well-structured system that meets both technical expectations and stakeholder needs.

COMMUNICATION PLAN

STAKEHOLDERS

- CLIENT: DR RHYS TAGUE
- SUPERVISOR: JACOB DESMOND
- TEAM MEMBERS: AHNAF, JONATHAN, SEANIYA, SHAURYA.

Channel	**Purpose**	**Frequency**
Email (Client)	Scheduling meetings, sending updates, sharing prototypes.	As needed (especially before Week 6).
Weekly Supervisor Meetings	Technical guidance, scenario validation, progress review.	Weekly.
Team Meetings	Task allocation, integration planning, debugging.	2–3 times per week.
Teams Chat/WhatsApp	Daily coordination, quick questions, code sharing.	Daily.
GitHub Repository	Version control, code reviews, issues tracking, documentation, scenario integration.	Continuous (daily commits recommended)
Prototype Workshop (Week 6)	Demonstrate early scenarios + harness.	Once (mandatory).

REPORTING STRUCTURE

- Weekly progress report:
- Iteration demos in week 9 and week 12 with sponsor confirmation
- Handover & completion report in week 14
- Final presentation of project in week 15

Handover Plan

When the project is finished, the team will provide the sponsor (robot researchers and aged-care supporters) with a fully working, ready-to-use PyBullet-based assessment harness. With just the supplied files and instructions, the handover guarantees that researchers without any prior experience with PyBullet may install, run, and provide assessment reports right away.

The documentation will include the following:

- Steps to install and run the simulations on pybullet
- Full explanation of every scenario with screenshots of expected results.
- Steps to change the parameters using only single line commands
- Troubleshooting section with common issues and its solutions

The team will also provide a presentation of the project after final handover to explain all the scenario outcomes and simulation working.

Handover Confirmation

By the end of week 14, the project will be handed over to the sponsor (Dr Rhys Tague), and within a week the presentation will be held. The handover will include all the python files for simulation, and a documentation that can be used to execute this simulation without any prior knowledge in pybullet.

Project Confirmation

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Sponsor

Dr Rhys Tague

Appendices

Appendix A (All scenarios provided by the sponsor)

ID	Scenario	Feasibility
S1	Patient engagement envelope (vertical reach map)	High (core)
S2	Bed ends reach (head vs foot zones)	High (core)
S4	Unreachable goal handling (early reject vs late failure)	High (core)
S7	Robot base placement comparison (placement trade-off study)	High (core)
S8	Time pressure vs safety trade-off (speed-limit study)	High (core)
S3	Dynamic goal update (patient/target shift)	Medium (core)
S5	Arm-region reach without collision	Low-Medium (extension)
S6	Reinforcement learning (optional stretch)	Low (stretch only)

Appendix B

PyBullet Implementation Notes

- Patient proxy: torso box + two arm cylinders anchored to bed frame
- Room layout: named bounding boxes (head/mid/foot; left/right side)
- Safety rules: minimum distance thresholds; collisions = hard failure
- Targets: poses relative to bed frame (success = end-effector distance + dwell time)
- Repeatability: deterministic seeds for all batch runs

Recommended Semester Scope

- **Core (must deliver)**: S1, S2, S4, S7 and S8 + complete evaluation harness (runner, logging, metrics, reporting).
- **Choose one extension**: S3 (Dynamic goal update) or S5 (Arm-region reach without collision).
- **Stretch only**: S6 (Reinforcement learning – optional, not required).